

NORWOOD, MA, USA

WMO: 725098

Lat: 42.191N

Lon: 71.174W

Elev: 15

StdP: 101.14

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 54704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-12.8	-23.9	0.4	-12.8	-21.5	0.5	-10.6	11.0	-1.0	9.8	1.2	1.9	300	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.8	23.1	31.3	22.4	29.6	21.5	24.9	30.3	24.0	28.6	23.2	27.2	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.0	27.1	22.6	17.3	26.4	22.0	16.7	25.9	75.7	30.3	72.1	28.7	68.8	27.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.1	7.2		-20.5	35.2	4.0	1.6	-23.4	36.4	-25.7	37.4	-27.9	38.3	-30.8	39.5
			DB	-20.9	26.2	3.9	1.0	-23.7	27.0	-26.0	27.6	-28.2	28.1	-31.0	28.9
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.4	-1.2	2.7	9.0	14.4	19.2	22.4	21.4	17.3	11.1	5.8	0.9
	DBStd	9.70	5.98	5.41	4.98	4.55	4.32	3.89	3.32	3.38	4.14	4.57	4.70	5.24
	HDD10.0	1482	385	315	232	72	7	0	0	0	1	41	141	287
	HDD18.3	3359	642	547	484	283	139	35	5	10	67	230	374	542
	CDD10.0	1525	1	2	7	41	143	277	384	354	221	74	17	4
	CDD18.3	359	0	0	0	2	16	62	131	106	37	5	0	0
	CDH23.3	3756	0	0	4	46	229	648	1365	1076	348	39	1	0
	CDH26.7	1274	0	0	1	15	74	213	517	358	92	4	0	0
(14) Wind	WSAvg	2.8	3.1	3.2	3.5	3.4	2.9	2.7	2.5	2.2	2.2	2.6	2.8	2.9
(15) Precipitation	PrecAvg	1206	98	87	118	103	95	98	86	98	96	111	100	114
	PrecMax	1552	183	206	390	229	245	335	198	201	237	339	169	219
	PrecMin	829	40	33	8	31	30	6	36	21	13	3	26	37
	PrecStd	195	35	39	75	49	48	85	38	47	55	75	39	48
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.5	17.0	21.8	28.8	32.2	33.8	35.2	34.2	32.4	27.0	21.3	18.4
		MCWB	14.1	12.3	13.3	16.8	20.1	23.6	25.0	24.6	23.2	19.2	16.5	14.7
	2%	DB	12.8	12.7	16.9	23.1	28.7	31.2	33.0	32.1	29.1	24.0	18.7	14.9
		MCWB	9.9	9.5	10.8	14.0	19.2	22.5	23.5	23.2	21.3	18.6	15.1	12.4
	5%	DB	9.1	9.8	13.3	19.9	25.8	28.9	31.4	30.3	27.2	21.9	16.5	12.1
		MCWB	6.4	6.3	8.5	12.7	17.3	20.7	22.7	22.2	20.5	17.5	12.9	9.5
	10%	DB	6.3	7.1	10.8	17.4	22.9	27.3	29.6	28.6	25.0	19.5	14.1	9.2
		MCWB	3.6	3.8	6.6	11.4	16.0	20.1	21.8	21.4	19.5	15.4	10.8	6.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	14.0	15.0	18.4	22.0	25.0	26.2	25.7	24.5	21.9	18.3	16.1
		MCDB	16.5	15.9	19.1	25.6	29.2	30.7	32.5	32.1	28.6	24.6	20.2	17.4
	2%	WB	10.6	10.0	12.3	16.4	20.5	23.7	25.1	24.7	23.1	19.9	15.9	12.9
		MCDB	11.9	12.3	16.0	21.5	26.6	29.1	30.9	29.7	26.5	22.6	17.7	14.9
	5%	WB	6.7	6.7	9.4	14.4	19.0	22.6	24.1	23.9	22.1	18.0	13.7	10.0
		MCDB	8.9	9.0	12.4	18.0	23.7	27.1	28.9	27.9	25.6	20.9	15.8	11.7
	10%	WB	3.9	4.0	6.8	12.3	17.3	21.4	23.2	22.9	21.1	16.2	11.6	7.1
		MCDB	5.9	6.4	9.9	16.0	21.2	25.3	27.3	26.6	24.0	19.0	13.6	9.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.8	10.5	11.7	12.3	11.9	12.1	12.4	13.0	12.5	11.2	9.8
		MCDBR	12.9	13.9	14.9	16.4	16.8	14.5	13.7	13.8	14.3	14.8	13.7	12.5
		MCWBR	11.0	10.7	10.0	8.8	8.3	7.0	5.9	6.3	7.8	9.7	10.1	11.0
	5% WB	MCDBR	12.6	12.6	13.6	13.5	14.3	12.3	11.5	11.1	11.1	12.2	11.7	11.8
		MCWBR	11.3	10.9	10.6	9.0	7.9	6.5	5.6	5.7	6.9	9.2	10.0	11.2
(40) Clear-Sky Solar Irradiance	taub		0.292	0.302	0.327	0.371	0.408	0.453	0.473	0.435	0.374	0.334	0.310	0.294
	taud		2.496	2.482	2.473	2.381	2.314	2.221	2.170	2.265	2.413	2.537	2.567	2.563
	Ebn at Noon		862	909	922	898	868	827	804	826	858	860	837	832
	Edh at Noon		72	86	98	115	127	140	145	128	103	81	68	63
(44) All-Sky Solar Radiation	RadAvg		1.80	2.67	3.67	4.71	5.22	5.66	5.93	5.33	4.30	2.86	1.96	1.49
	RadStd		0.12	0.23	0.26	0.50	0.57	0.54	0.35	0.25	0.30	0.25	0.15	0.16

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	N/A
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+44

Nomenclature: See separate page